

Project Title: Constraint-Aware Fitness Planner

A Personalized Diet and Workout Planning Agent for Users with Health, Lifestyle, and Adherence Constraints

Team Members: Shiwei Jiang

Track B: Applied Agent Experience

Project Summary

Constraint-Aware Fitness Planner is an interactive agentic planning system that helps users create personalized diet and workout strategies for fat loss, muscle gain, or body recomposition while accounting for real-life restrictions such as food sensitivities, chronic symptoms, schedule limitations, budget, exercise access, and adherence challenges. Unlike a single-turn LLM application, this system uses a multi-agent, stateful workflow to track user-specific constraints. It collects user information, identifies hard constraints and conflicts, generates an initial plan, reviews that plan for realism and safety, and revises it when the user reports new limitations or difficulty following the plan. The project is intentionally scoped as an autonomous decision-support and replanning experience rather than a medical tool. Its purpose is to produce plans that are more realistic, personalized, explainable, and sustainable than typical generic online fitness advice.

Target User / Stakeholder / Audience

The primary user is an individual trying to lose fat, gain muscle, or improve body composition, but whose planning needs are shaped by multiple constraints, including health-related restrictions, food intolerances, chronic symptoms, time limitations, budget, exercise access, and consistency challenges.

The secondary stakeholder is the future evaluator or reviewer of the system, who needs to see that the system produces recommendations that are not only personalized, but also bounded, explainable, and governable.

Problem Statement

Many diet and workout plans fail not because users lack motivation, but because generic recommendations optimize only for broad goals like “weight loss” or “muscle gain” and ignore the interaction among personal constraints. A meal or workout pattern that is effective for a healthy average user may be inappropriate for someone with food sensitivities, chronic urticaria, digestive issues, poor recovery, limited cooking time, or inconsistent scheduling. As a result, users often receive plans that look good in theory but are unsafe, unrealistic, or difficult to follow in practice. This project addresses that gap by building a planning system that reasons over goal–constraint conflicts, not just idealized fitness targets.

Why this is an Agentic Problem

This is an agentic problem because the system must demonstrate a dynamic process characterized by autonomous decision-making and multi-step reasoning. It goes beyond merely

generating a static plan; instead, it actively navigates complex constraint conflicts through an "Observe-Think-Act" loop. It must: collect structured user information; identify hard constraints and soft preferences; detect conflicts among goals, restrictions, and lifestyle realities; generate a candidate plan; review that plan for realism, safety, and adherence risk; and revise it when the user reports new constraints or failed execution. The value of the system comes from multi-step decision-making, role-based handoffs, stateful adaptation, and iterative replanning, not from a one-shot answer. When user feedback indicates execution failure or when environmental conditions change, the system, much like a human coach, updates its understanding and adjusts its subsequent strategies, rather than simply offering static, fixed responses.

Initial Architecture Concept

The system will use a 4-component role-based workflow with shared session state.

Component 1: Intake & Profile Builder

Collects user goals, body metrics, food preferences, restrictions, exercise context, budget, and schedule, and converts them into a structured profile.

Component 2: Constraint & Risk Checker

Analyzes the profile to identify hard constraints, soft preferences, missing critical information, and conflicts such as unrealistic targets, food–restriction mismatches, or workout plans that exceed schedule or recovery limits.

Component 3: Plan Composer

Generates a diet and workout strategy, weekly structure, and rationale based on the user profile and constraint report.

Component 4: Review & Adaptation Agent

Reviews the proposed plan for safety, realism, and adherence risk, then either approves it, requests revision, or adapts it later after user feedback such as “I could not follow this plan” or “this food caused a problem.”

Shared Session State

Stores the structured profile, constraint report, plan version history, review notes, and user feedback so that the system can support stateful replanning instead of restarting from scratch every turn.

This architecture is intentionally limited to four purposeful roles so that the system remains clear and feasible while still showing meaningful coordination and agentic behavior, where the extra complexity is justified rather than decorative.

Initial Success Criteria

For this project, initial success means:

1. **Personalization quality**

The generated plan reflects the user’s stated goal, food restrictions, schedule, exercise access, and major constraints rather than giving a generic template.

2. **Constraint handling**

The system identifies major conflicts or missing information before finalizing a plan.

3. **Safety boundary compliance**

The system avoids diagnosis and treatment advice and escalates when requests exceed the project's safe scope.

4. **Adherence realism**

The plan is not only theoretically effective but also plausible for the user to follow.

5. **Replanning ability**

When the user reports failure, discomfort, or changed constraints, the system can revise the plan coherently instead of regenerating an unrelated answer.

6. **Explainability**

The system gives brief rationale for key diet and workout decisions.

Scope Boundaries

In scope

- Personalized fitness planning for fat loss, muscle gain, or recomposition.
- Constraint-aware diet strategy and workout strategy generation.
- Goal–constraint conflict detection.
- Schedule-aware and adherence-aware recommendations.
- Plan revision after user feedback or failed adherence.
- Safety disclaimers and escalation behavior when the request exceeds safe planning scope.

Out of scope

- Medical diagnosis.
- Treatment recommendations.
- Medication advice.
- Clinical nutrition prescription.
- Real-time biomarker monitoring.
- Automatic calorie counting from images.
- Long-term disease prediction.
- Any claim that the system replaces physicians, dietitians, or licensed professionals.

Initial Risks or Concerns

1. **Medical-overreach risk**

Users may ask for diagnosis-like advice or interpret suggestions as medical guidance.

2. **Hallucinated confidence risk**

The system may sound overly certain even when user information is incomplete.

3. **Constraint-miss risk**

The system may overlook a user-specific restriction and generate an unsuitable recommendation.

4. **Adherence optimism risk**

The plan may be technically “good” but too difficult to sustain in real life.

5. **Ambiguity risk**

Inputs like “I have stomach issues” or “I react badly to some foods” may be too vague unless the system asks follow-up questions.

6. **Evaluation difficulty risk**

Unlike a math problem, “best plan” is not fully objective, so evaluation must focus on structured expected behavior, conflict detection, safety boundaries, and revision quality.

Short Team Plan

Problem framing, architecture design, prototype implementation, evaluation design, testing, failure analysis, documentation, and final presentation. To keep the project feasible, the implementation will focus on the core interaction workflow rather than broad feature expansion.

Track B Addition: Initial User Journey / Interaction Map

Roles / Actors in the Experience

- User
- Intake & Profile Builder
- Constraint & Risk Checker
- Plan Composer
- Review & Adaptation Agent
- Shared Session State

Initial User Journey

1. User enters personal goal, body metrics, preferences, restrictions, time availability, and workout context.
2. Intake & Profile Builder structures the information.
3. The Constraint and Risk Checker then identifies hard constraints, major conflicts, missing critical information, and caution flags.
4. If information is insufficient, the system asks follow-up questions before proceeding.
5. Once information is enough, plan Composer creates an initial diet and workout plan.
6. Review & Adaptation Agent checks the plan for safety, realism, and adherence risk and either approves it, requests a revision, or returns the reviewed version plus rationale and any caution notes.
7. The user then tries the plan in the real world.
8. The user later provides feedback such as adherence difficulties, discomfort, or changes in schedule or constraints.
9. The Review & Adaptation Agent updates the plan while preserving prior context and explaining the rationale for revisions.

10. This feedback and revision loop continues iteratively until a satisfactory and feasible plan is reached.

Where “agency” lives in the design

Agency lives in the system’s ability to decide when more information is needed, identify conflicts, determine whether a plan is ready or requires revision, preserve relevant context across turns, and adapt the plan after feedback rather than simply answering once. The system is not only generating content; it is making bounded workflow decisions about clarification, conflict handling, revision, and escalation.

Multi-agent System Canvas

1. System Intent

Use Case / Purpose

Provide a personalized, constraint-aware fitness planning experience for users pursuing fat loss, muscle gain, or body recomposition, while accounting for health-related restrictions, food sensitivities, time limits, exercise context, and adherence challenges.

Task Slice / Goal

Generate a realistic diet-and-workout planning package that: matches the user’s stated fitness goal, respects hard constraints and important preferences, flags major conflicts or missing information, and can be revised after feedback or failure to follow the plan.

Users / Stakeholders

Requester: individual end user seeking a personalized fitness plan

Reviewer: course evaluator / project reviewer

Downstream actor: the same user, who attempts to follow the plan and returns feedback for revision.

2. Bounded Behavior

Inputs / Outputs

Inputs:

User goal, height/weight/body context, schedule, dietary preferences, food restrictions, health-related restrictions, exercise access, budget, adherence concerns, follow-up clarifications, later feedback

Outputs:

Structured user profile, constraint report, initial fitness plan, reviewed plan with rationale, revised plan after feedback, boundary warnings or escalation messages when necessary

Tools / Permissions

Allow:

Structured intake form or schema, lightweight rule layer for conflicts, LLM-based planning workflow, stored session state, plan templates

Deny:

Diagnosis tools, medication recommendation tools, automatic medical interpretation, direct clinical claims, external action-taking on behalf of users

Autonomy / Human Checkpoints**Autonomous:**

Profile structuring, conflict detection, initial planning, plan review, revision suggestions

Human checkpoint:

User confirms or corrects profile information

User decides whether to accept the plan

User provides feedback after trying the plan

User must consult qualified professionals for diagnosis/treatment matters

Operating Rules / Policies

Hard rules: Do not diagnose diseases. Do not recommend medication changes. Do not ignore stated restrictions. Ask follow-up questions when critical information is missing.

Soft rules: Prefer sustainable plans over aggressive plans. Explain trade-offs briefly. Flag ambiguity rather than pretending certainty.

Project policy: This is a planning assistant, not a medical or clinical system

3. Context and Evaluation**Memory / Context**

Session memory: structured user profile, constraint report, plan versions, review notes, feedback history

Persistent memory: none beyond the project session unless explicitly enabled later

Privacy rule: do not retain sensitive user information beyond what is necessary for the current interaction in the prototype

Failure Conditions / Escalations

Missing info: pause and ask targeted follow-up questions

Medical-overreach request: refuse diagnosis/treatment and recommend professional consultation

High-risk plan detected: downgrade intensity or escalate with caution

Constraint conflict unresolved: present limitation and ask for prioritization

Low confidence due to vague user input: say what is missing rather than guessing

Success Criteria

Task success: user receives a plan that matches the stated goal and respects major constraints

Constraint success: plan avoids obvious conflicts with stated restrictions

Safety success: system stays within non-diagnostic scope

Adaptation success: revised plan responds coherently to feedback

Quality success: rationale is understandable and specific enough for the user to act on

Refine the Specification

Tighten scope if medical complexity starts to dominate.
Adjust permissions if the system begins to overclaim authority.
Clarify human checkpoints if the system's safe stopping conditions are unclear.

Agent Canvas for Each Agent / Component

Agent Canvas 1 — Intake & Profile Builder

System Intent

Transform messy user input into a structured planning profile.

Use Case / Purpose

Collect relevant information about goals, metrics, lifestyle, diet, exercise context, restrictions, and preferences so later components can reason over it consistently.

Task Slice / Goal

Produce a normalized **User Profile** that includes goal, major constraints, preferences, schedule, and context.

Users / Stakeholders

Primary user: end user

Downstream component: Constraint & Risk Checker

Inputs / Outputs

Input: free-text and structured intake responses

Output: normalized user profile object

Tools / Permissions

Can use intake schema, follow-up question prompts, structured field extraction

Cannot give final plan or medical interpretation

Autonomy / Human Checkpoints

Autonomously organizes user information

Asks the user follow-up questions if key profile fields are missing

Operating Rules / Policies

Do not infer medical details that the user did not provide

Separate **hard constraints** from **soft preferences**

Mark unknowns explicitly

Memory / Context

Stores the current profile in session state

Failure Conditions / Escalations

If user input is too vague or contradictory, request clarification before moving on

Success Criteria

Profile is complete enough for downstream planning and clearly separates goals, restrictions, and preferences

Agent Canvas 2 — Constraint & Risk Checker

System Intent

Detect conflicts before a plan is finalized.

Use Case / Purpose

Evaluate whether the user's goal, restrictions, preferences, schedule, and resources create any important incompatibilities or risks.

Task Slice / Goal

Produce a **Constraint Report** listing hard constraints, soft preferences, detected conflicts, missing critical information, and preliminary caution flags.

Users / Stakeholders

Primary downstream users: Plan Composer and Review & Adaptation Agent

Inputs / Outputs

Input: structured user profile

Output: constraint report

Tools / Permissions

May use lightweight rule checks, conflict-detection prompts, and predefined planning heuristics
May not make diagnosis or treatment claims

Autonomy / Human Checkpoints

Autonomously identifies conflicts

If conflict cannot be resolved, asks the user to prioritize constraints

Operating Rules / Policies

Always respect user-stated restrictions

Flag aggressive targets or unrealistic combinations

Prefer “ask for clarification” over guessing

Memory / Context

Reads profile from session state and writes back a constraint report

Failure Conditions / Escalations

If medical issues or symptom-related restrictions exceed safe interpretation scope, issue boundary warning and restrict recommendations

Success Criteria

Major conflicts are caught before plan generation, and unresolved ambiguity is surfaced clearly

Agent Canvas 3 — Plan Composer

System Intent

Generate the first-pass personalized fitness plan.

Use Case / Purpose

Create a diet-and-workout strategy tailored to the user's goal and bounded by the constraint report.

Task Slice / Goal

Produce an **Initial Plan Package** including nutrition strategy, workout strategy, weekly structure, and brief rationale.

Users / Stakeholders

Primary user: end user

Reviewer: Review & Adaptation Agent

Inputs / Outputs

Input: user profile + constraint report

Output: initial plan package

Tools / Permissions

May use templates, planning heuristics, and LLM-based reasoning over constraints

May not prescribe medication, diagnose conditions, or claim clinical certainty

Autonomy / Human Checkpoints

Autonomously generates the first draft plan

Human later decides whether to accept or revise

Operating Rules / Policies

Every major recommendation should map back to at least one user goal or constraint

Prefer sustainable plans over maximally aggressive plans

Include brief rationale for key decisions

Memory / Context

Reads profile and constraint report, stores plan version in session state

Failure Conditions / Escalations

If constraints are too severe or too underspecified, produce a limited plan plus clarification or escalation note

Success Criteria

Plan is personalized, coherent, and evidently shaped by user-specific constraints rather than generic advice

Agent Canvas 4 — Review & Adaptation Agent

System Intent

Review and revise the plan for safety, realism, and adherence.

Use Case / Purpose

Critique the initial plan, identify hidden issues, and update it after user feedback or plan failure.

Task Slice / Goal

Produce either an approved reviewed plan, a request for revision, or a revised plan based on later user feedback.

Users / Stakeholders

Primary user: end user

Downstream effect: accepted final plan or revised plan

Inputs / Outputs

Input: user profile, constraint report, current plan, later user feedback

Output: reviewed plan, revision notes, or adapted plan

Tools / Permissions

May use review rubric, adherence heuristics, and revision logic

May not exceed project safety boundaries

Autonomy / Human Checkpoints

Autonomously critiques the plan

Prompts the user for feedback after attempted execution

Allows the user to override or decline recommendations

Operating Rules / Policies

Check for: unrealistic intensity, hidden conflict with restrictions, overcomplication, poor adherence fit, and unsafe boundary crossing

If the plan exceeds safe scope, stop and escalate instead of improvising

Memory / Context

Reads previous plan versions and user feedback from session state, then stores revised versions and reasons for changes

Failure Conditions / Escalations

If the user reports symptom-like issues or asks for disease-specific interpretation, the system must stop short of diagnosis and recommend professional guidance

Success Criteria

The system can explain what changed, why it changed, and how the revised plan better fits the user's situation

Appendix: AI Usage Log

Tool Name and Version: Chatgpt instant

1. Scoping and Brainstorming

- What I used it for: Refining the project topic and narrowing down the "Track B: Applied Agent Experience" focus.
- Exact Prompt/Task: "Could you help me brainstorm a few more ideas? Specifically, I'm looking for concepts related to: - "Personalized weight loss or muscle-building plans tailored to individual body constitution" - A Travel Decision Support Agent (designed to help solo travelers compare itinerary options while adhering to budget, time, and risk constraints) - A Training Assistant to facilitate the onboarding of both new and existing employees - Digital Twins and the training of embodied AI agents - Robotic sensor systems In addition to these, please feel free to suggest a few other ideas as well; however, I would prefer to avoid topics related to finance, as I lack the necessary prior knowledge in that domain. Finally, please let me know which of these ideas you think best align with the project requirements and would be the most suitable to pursue." "Give me some brainstorm ideas about what I can do... I want to focus on a Fitness Personalized Planner that handles complex constraints like chronic urticaria and specific diets." "
- Manual Changes: I manually narrowed the scope to focus specifically on conflict resolution (e.g., goal vs. health) rather than just "generating a workout," as per the specific requirement to show "Agency."
- Verification: Verified that the "Constraint-Aware" approach met the course definition of an agentic system (multi-step reasoning and state management).

2. Architecture and Role Definition

- What I used it for: Drafting the multi-agent workflow and defining specific responsibilities for each agent.
- Exact Prompt/Task: "According to the last two answers in the PDF, write Phase 1 as a final submission document... ensuring it meets the rubric requirements for role-based handoffs and closed-loop feedback."
- Manual Changes: I reorganized the agent roles to ensure the "Adaptive Replanner" was a distinct entity from the "Plan Generator" to satisfy the rubric's requirement for iterative feedback loops.
- Verification: Cross-referenced the agent roles with the Phase 1 Rubric to ensure "Safety Reviewer" and "Constraint Checker" provided the necessary governance.

3. Safety and Risk Assessment

- What I used it for: Identifying potential failure cases and project risks.
- Exact Prompt/Task: "Identify initial risks and scope boundaries for this fitness agent."
- Manual Changes: I specifically added the "Medical Boundary" risk and the "Non-medical" constraint limitation to ensure the project stays within safe, ethical bounds for an AI system.
- Verification: Verified that the proposed mitigations (disclaimers and rule-based escalation) are standard industry practices for health-related AI agents.

4. Refine my explanation

- What I used it for: refine “why agentic problem”
- Exact Prompt/Task: Add the explanation of “Observe-Think-Act” loop into the “why agentic problem” part.
- Manual Changes: I add my own understanding and content from the lecture. I deleted “The project justifies a distinct architecture with meaningful roles, components, branches, or handoffs” from the response.
- Verification: it aligns with the course slide

I have reviewed all AI-generated content for accuracy. The final design decisions regarding the system architecture and the specific constraint-handling logic are my own.